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RESEARCH ARTICLE

Peduncle Detection of Ripe Strawberry to Localize Picking Point Using DF-Mask R-CNN and Monocular Depth

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ABSTRACT Accurate localization of picking points and depth estimation is critical for implementing a robotic strawberry harvesting system. Due to the delicate nature of strawberries, harvesting must be performed without bruising or damage, typically by grasping and cutting the peduncle of the ripe strawberry. However, accurately detecting and localizing the thin peduncle in a cluttered environment is a significant challenge. This study proposed depth fused Mask R-CNN (DF-Mask R-CNN), which integrates depth information of the scene with the RGB image to enhance the detection, localization, and segmentation of strawberries and their peduncles in a greenhouse environment. To generate a dense depth map, a cutting-edge monocular depth estimator, ZoeDepth was used. The proposed DF-Mask R-CNN with ResNet101-FPN exhibited superior instance segmentation performance, with an overall mAP of 81.9%, with mAP_{small} at 33.3%, mAP_{medium} at 78.79%, mAP_{large} at 88.8 and AP^{IOU=0.5} at 98.1%. In tests with 300 ripe strawberry samples, the method demonstrated a robust picking point detection, with a mean absolute error and root mean square error of 1.98 cm and 2.12 cm, respectively. These results highlight the effectiveness of the DF-Mask R-CNN model combined with the ZoeDepth estimator in enhancing the detection, localization, and segmentation of strawberries and their peduncles. This approach enables precise picking point localization and depth estimation for efficient vision systems for robotic strawberry harvesting.

INDEX TERMS Instance segmentation, Mask R-CNN, monocular depth estimation, peduncle detection, picking point estimation, ripeness detection.

I. INTRODUCTION

Strawberries are among the most highly valued berries cultivated globally [1]. Over the past decade (2010-2019), the global strawberry production had an average growth rate

of 3.2% per year [2]. As global production for strawberries increases, finding ways to increase production efficiency is essential. Strawberry production faces a substantial labor shortage due to labor-intensive workloads and has economic challenges due to rising labor cost. Among different strawberry production stages, harvesting involves extensive work hours and represents the highest operational expense.

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Harvesting consists of a quarter of the total working hours required, whereas the cost-related accounts for 40-60% of overall labor costs [3], [4]. Therefore, an autonomous strawberry robotic harvester offers a potential solution for the scarcity of agricultural workers and escalating labor expenses [5]. While several prototype strawberry harvesting robots have been developed, detecting precise and accurate ripe strawberries and efficient harvesting in an on-field environment remains a significant challenge to their widespread adoption [6].

Autonomous harvesting robots utilize vision systems comprising RGB and depth cameras to determine different ripeness levels of strawberries and their location. Ripe strawberries are soft-skinned and prone to damage and bruise. This requires special care in the design of the robotics' end effector and its operation [7]. Robotic harvesters grasp and cut the peduncle associated with ripe strawberries in harvesting. This involves the identification of peduncles associated with ripe strawberries and then precisely estimating the picking point on the peduncle for effective harvesting [8]. The vision system performs these functions using object detection and localization algorithms. Historically, image-processing algorithms employed basic image-processing operations, such as thresholding, edge detection, region growth, and erosion, to extract features such as morphology, chromatic, and surface properties. Furthermore, these features were input to machine learning algorithms like support vector machine (SVM), Random Forest (RF), and Artificial Neural Networks (ANN) to identify strawberry attributes such as shape, size, ripeness level, and disease detection [9], [10], [11]. However, feature extraction methods carried out in these primitive algorithms were hand-crafted and restricted to controlled environments tailored to specific breeds [6]. In the greenhouse detection of ripe strawberries, machine learning models face different plant structural and environmental challenges [12], posing difficulty in generalizing, which leads to the exploration and application of deep learning technology.

The advent of deep learning (DL), particularly convolution neural networks (CNN), enhanced object detection models, which are pivotal for tasks such as identifying, localizing, and assessing strawberry ripeness levels under complex environments. DL-based object detection methods are categorized into one-stage detectors such as You Only Look Once (YOLO) and Single Shot Detector (SSD) [13], [14], and two-stage detectors like Region-based Convolution Neural Network (R-CNN) and its variant, Faster R-CNN. These detectors identify different ripeness levels and localize the strawberries using bounding boxes. Recent advancements have focused on improving model performance and decreasing computational requirements. In the study by An et al. [15], an improved YOLOX model (SDNet model) was developed, incorporating a self-designed feature map and SIOU loss function to enhance detection speed and performance notably. Similarly, Tang et al. [16] integrated a lightweight RepGhost model into the YOLOv7-tiny

framework to address challenges in detecting small, overlapping strawberries and decreasing computational requirements. While the bounding box localizes the identified strawberry, it cannot correctly estimate its total visibility and pose. Proper morphological information, such as visible shapes and poses, is essential to calculate and estimate the peduncle and picking point. The requirement has given rise to instance segmentation, an advanced Mask R-CNN model that generates pixel-level masks on the scene. Yu et al. [6] used the Mask R-CNN to successfully identify and generate segmentation masks for isolated and overlapping strawberries, achieving precision, recall, and mean intersection over union (IOU) of 95.78%, 95.41%, and 89.85%, respectively. The instance segmentation segregates isolated and partially occluded strawberries.

Most often, studies focused on harvesting ripe strawberries evaluate the picking point on the peduncle, which is assessed based on the strawberry's overall shape [16], [17]. The principal axis of the strawberry is estimated, then the likely location of the peduncle on the axis is determined, and finally, the picking point is estimated. The shape-based approach has inherent limitations because the accuracy of the picking point directly depends on the accuracy of the perceived shape and visibility. Moreover, in a real-world environment, strawberries are partially or severely occluded, resulting in inaccuracies in estimating the picking point.

Direct detection of peduncles associated with strawberries can potentially improve the effectiveness of harvesting operations. However, peduncle detection poses challenges due to its thin structure, small size, and similar colour to the background. Moreover, the difficulty increases because of cluttered environments and variable lighting conditions (as illustrated in Figure 1). These issues lead us to focus on these underexplored problems and devise a method for detecting strawberries with different ripeness levels and their peduncles. To address these challenges, we introduced additional depth information to the conventional RGB-based Mask R-CNN that brings spatial awareness for effective detection, localization and segmentation performance. Traditional sensors, like LiDAR and stereo cameras, can acquire in-depth information. These sensors often provide precise depth measurements for big objects and objects within the defined specified range. However, depth inaccuracies occur for small objects and objects located out of the specified range [18]. To overcome these issues, we explored the software-based solution, i.e., monocular depth estimators.

Monocular depth estimator is a state-of-the-art deep learning algorithm that can generate dense depth maps of scene from only a single RGB image. This has been widely used in autonomous driving and augmented reality. The study by Eigen et al. [19] initiated end-to-end learning of depth from monocular images. The enhanced network architecture, loss functions, and training strategies improving both accuracy and efficiency, were carried out in subsequent research. The key advancements include encoder-decoder structures,



FIGURE 1. Strawberries and their associated peduncles. in (a) dense fruition, (b) unstructured environments.

skip connections, and multi-scale feature fusion, as demonstrated in the works of Laina et al. [20] and Fu et al. [21]. Self-supervised learning paradigms, pioneered by Garg et al. [22] and refined by Zhou et al. [23], enabled training on unlabeled video sequences, addressing the shortage of ground truth depth data.

Therefore, this study aims to investigate the integration of dense depth maps in Mask R-CNN to evaluate its effectiveness and efficiency in detecting strawberries and their peduncles, which helps the autonomous harvesting system by estimating accurate picking points. To achieve this aim, we propose the following three-fold approach:

- 1) Create a comprehensive dataset comprising strawberries at various ripeness levels along with their peduncles
- 2) Employ a state-of-the-art monocular depth estimation model to generate dense depth maps for incorporating multimodal input (RGB-D) to the conventional Mask R-CNN
- 3) Design and implement the DF-Mask R-CNN, which integrates depth information along with RGB imagery to enhance instance segmentation generating for precise and accurate contour of strawberry and its peduncle.

The rest of the paper is structured as follows: Section II presents the overall system overview, image acquisition and processing, followed by the generation of segmentation masks for strawberries and their peduncle—an introduction to the monocular depth estimator necessary to output dense depth information. After proper localization and obtaining the mask of the strawberry and its peduncle, the algorithm for estimation of the picking point was done. Further on, the experimental environment, parameter setting, and evaluation metrics are explained. Section III evaluated the performance comparison of conventional Mask R-CNN and DF-Mask R-CNN on standard metrics. In section IV, we compared the performance of the DF-Mask R-CNN model on various conditions and with relevant research studies. Finally, section V concludes the paper with a summary of the findings and implications.

II. MATERIAL AND METHODS

A. SYSTEM OVERVIEW

Figure 2 illustrates a flow diagram of the system developed to identify the ripe strawberry and its peduncle and estimate the picking point. The system's main blocks were dataset acquisition, depth map generation, multimodal input fusion, enhanced instance segmentation, and picking

point evaluation. Dataset acquisition consists of the primary sensor, Intel RealSense LiDAR L515 (Table 1), which acquired the scene information in the form of RGB and depth information. Enhanced instance segmentation generates detailed contours of the detected strawberry and its peduncle. Finally, the shape information of the detected peduncle was used to estimate the harvesting point on the peduncle precisely; otherwise, the shape information of the visibly detected strawberry was utilized to evaluate the picking point on the absence of the visible peduncle. During picking point estimation, the depth of the picking point on the peduncle of the ripe strawberry from the camera was evaluated based on camera intrinsic information obtained from LiDAR.

B. DATASET ACQUISITION AND PREPARATION

1) ACQUISITION OF IMAGES

The experiment was conducted at the Smart Farm System Laboratory, Gyeongsang National University, Jinju, Republic of Korea. Strawberries of the Seolhyang variety were cultivated on a raised bed with five rows spaced one meter apart. Images were collected during two harvesting seasons: the first on December 4, 2023, at the BBCH (Biologische Bundesanstalt, Bundessortenamt und Chemische Industrie) phenological growth stage 85, and the second on February 14, 2024, at stage 89 [24]. The LiDAR camera was mounted on a manual mobile platform at the height of 0.8 m to capture images of strawberries hanging from the raised beds, with the camera-to-fruit distance ranging from 30 to 80 cm. The distance allowed us to acquire close-up images of strawberries as well as strawberries with the plant canopy.

Data collection occurred between 9:00 AM and 5:00 PM during different periods (morning and afternoon) to capture images under varying illumination (sunny and cloudy conditions) to enhance the generalizability of deep learning models. The dataset includes various instances of occlusions on strawberries and peduncles caused by stems, leaves, and overlapping strawberries in a greenhouse environment, as illustrated in Figure 3.

2) DATASET PREPARATION

The initial dataset preparation stage involved annotating strawberries based on their maturity levels and peduncles of strawberries. Strawberries were classified into three maturity levels: immature, transition, and mature, as a reference to a study by Tamrakar et al. [12]. The polygon annotation was done using Label Studio software, as shown in Figure 4. Manual annotations are labor-intensive, requiring significant effort and time to label the contour of a strawberry and its peduncle. Therefore, the semi-automatic annotation feature available in Label Studio was utilized by configuring the “Segment Anything” machine learning (ML) model in the backend. The setup allowed accelerated polygon annotation generation by simply labelling the target strawberry with rectangular bounding boxes. Strawberry annotation comprises

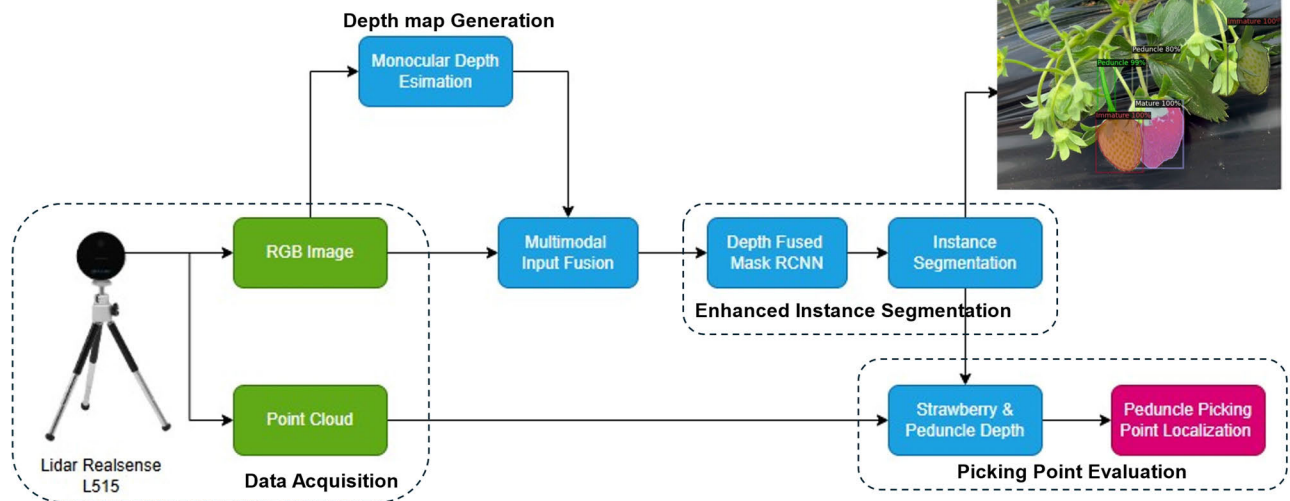


FIGURE 2. Flow diagram of the system for detecting picking points on the peduncle of ripened strawberry.

TABLE 1. Image acquisition sensor, specification of intel RealSense LiDAR L515.

Specifications	Details
Manufacturer and Mode	Intel RealSense Lidar L515
RGB channel resolution (pixels)	1920 x 1080 @ 30 fps
RGB channel field-of-view (FOV)	70° x 43° (±3°)
Depth channel resolution (pixels)	1024 x 768 @ 30 fps
Depth channel FOV	70° x 55° (±3°)
Working range (m)	25 cm to 9m

both fruit as well as calyx. Once semi-automatic annotation generated a satisfactory strawberry contour, the annotation was confirmed. However, semi-automatic annotation was often inadequate for the thin peduncles, necessitating manual annotation. All peduncles associated with different maturity levels were annotated to incorporate varying peduncle structures. Peduncle was annotated to their ramification points for detailed identification. During annotation, both the strawberry and its peduncle annotation overlapped with each other. The overlap allowed the detection model to learn the association between the strawberry and its peduncle, aiding in accurately detecting the peduncle associated with the strawberry. Examples of annotated images are shown in Figure 5.

The dataset consisting of strawberry images and respective classes is detailed in Table 2. BBCH85 and BBCH89 consist of 700 and 600 images. The BBCH89 dataset was used to train and validate the model, while BBCH85 was used for testing. The training dataset was divided into training and validation sets in an 80:20 ratio, meaning 560 images were used for training and 140 for validation. The validation set was used for cross-validation of the trained models.

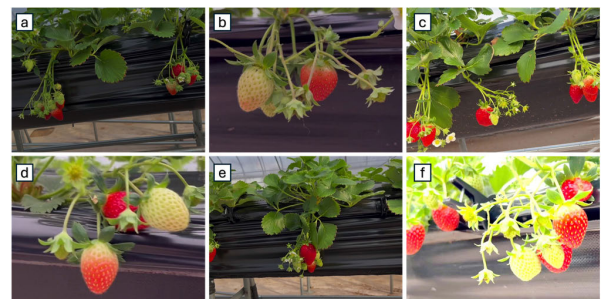


FIGURE 3. Difference instances of images sample of strawberries and peduncle (a) multiple strawberries and green foliage with clear peduncle under low lighting (b) close-up strawberry with lots of branching (c) strawberry and peduncle in unstructured environment (d) close-up strawberry with occlusion and ill pose (e) Strawberry with no visible peduncle (f) dense fruition under intense light.

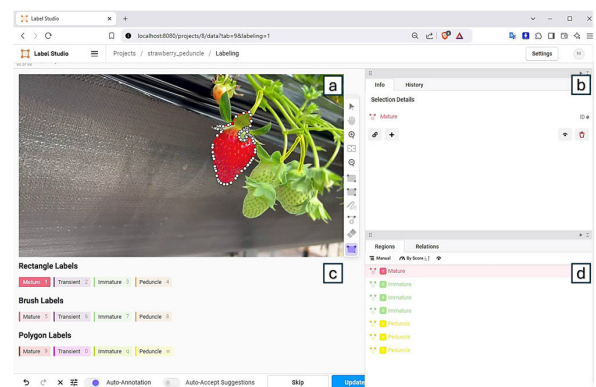


FIGURE 4. Label studio annotation software with semi-automatic annotation (a) Labeling window (b) Annotations details (c) Annotations label options (d) Annotations class details.

C. INSTANCE SEGMENTATION OF STRAWBERRIES AND PEDUNCLES USING DF-MASK R-CNN

Mask R-CNN [25], a well-known two-stage network was utilized to detect bounding boxes and perform instance

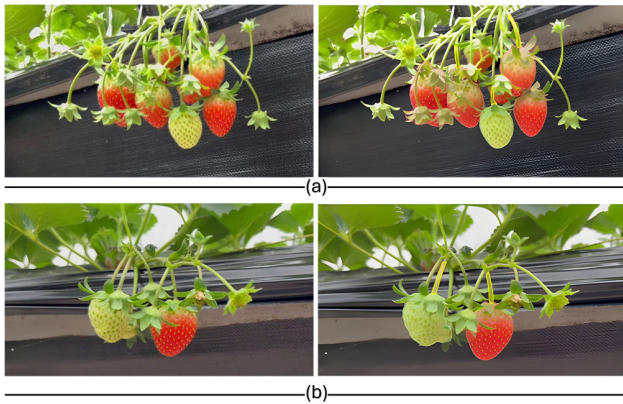


FIGURE 5. Annotation of images with four classes: (a) annotation with strawberries and their associate peduncle class, (b) peduncle annotated up to the branching point.

TABLE 2. Details of the strawberry image.

Dataset	Division	Image	Immature	Transition	Mature	Peduncle
BBCH89	Train	560	685	360	915	1060
	Valid	140	180	150	155	245
BBCH85	Test	600	-	-	-	-
	Total	1300	865	510	1070	1305

segmentation of the strawberry classes and peduncle. The Mask R-CNN model was implemented using the Detectron2 framework [26] as the baseline architecture. The model comprises three key components: the backbone, the region proposal network (RPN), and the region of interest (ROI) head.

The backbone network serves as the initial stage and is responsible for extracting feature maps from the input scene, which are essential for detecting and segmenting strawberries and their peduncles. This study used two variants of ResNet Feature Pyramid Network (FPN), a deep learning block, ResNet50-FPN and ResNet101-FPN as backbone networks. ResNet-FPN block consists of a stem block with 7×7 convolution layers and max pooling units, four stages of residual blocks with lateral connection, and an FPN block with a top-down pathway merging coarse semantic features and fine-grained spatial features. The Region Proposal Network (RPN) block employs 15 anchor boxes per location and output from FPN-generated feature maps to propose object candidates through 3×3 convolution (512-d feature encoding) and twin 1×1 convolution heads having objectness scoring and bounding box adjustment.

Once proposals are generated, ROI Align uses bilinear interpolation to extract 14×14 feature maps from proposal regions, eliminating quantization errors through continuous coordinate mapping. This preserves pixel-perfect alignment critical for mask accuracy. These representations are then processed by a multi-branch prediction network

for classification, bounding box regression, and mask prediction. The classification branch consists of two 1024-d fully connected layers that predict the ripeness and peduncle classes; the bounding box regression branch consists of two 1024-d fully connected layers that refine target localization; and the mask prediction branch consists of four 3×3 convolution layers, 2x up sampler, and 1×1 convolution layer to generates a pixel-level mask outlining the specific strawberry and peduncle shape. While the conventional Mask R-CNN relied solely on RGB images to identify and segment strawberries and their peduncles, this study, DF-Mask R-CNN, hypothesized that incorporating depth information into the framework can significantly enhance instance segmentation performance. The detailed schematic diagram of the DF-Mask R-CNN model is illustrated in Figure 6. This model takes RGB images and dense depth maps as input. Dense depth maps were generated from RGB images using the monocular depth estimator, ZoeDepth model. The depth map was fused with an RGB image at the stem layer, the initial stage in the ResNet model of the backbone. Originally, the ResNet stem layer could only take RGB images as an input. The stem layer was information to the RGB image. Modified stem aggregate RGB 3 channel input with dense depth channel obtained after 1×1 convolution. The depth fused image after the 4-channel RGBD image is shown in Figure 7. After the depth fusion in the input layer, the operation follows a process similar to Mask R-CNN for strawberry identification and segmentation.

D. MONOCULAR DEPTH ESTIMATION

DF-Mask R-CNN utilized a dense depth map of the scene to improve segmentation. To generate a dense depth map from the single RGB image, the study uses the monocular depth estimation ZoeDepth, sourced from GitHub repositories <https://github.com/isl-org/ZoeDepth> [27]. The ZoeDepth model uniquely integrates relative and metric depth estimation techniques, tackling the challenge of inferring 3D structures from 2D data without stereo or temporal information. It was pre-trained on multiple datasets using relative depth information, which helps it learn general depth relationships and improve generalization to unseen scenes. Subsequently, the model was fine-tuned on specific datasets with metric depth ground truth for accurate real-world scale measurements. Its architecture employs a lightweight head with a novel “metric bins module” for each domain. During inference, input images were automatically routed to the appropriate head using a latent classifier. This dual approach resulted in a versatile model demonstrating strong zero-shot transfer capabilities to unseen datasets from both indoor and outdoor environments while maintaining metric accuracy in specific scenarios. The Zoe model provides a comprehensive solution for various depth estimation applications from single RGB images. This demonstrates its significant potential in agriculture applications such as fruit size, shape, weight estimation and harvesting without using expensive depth sensors and high amounts of data processing.

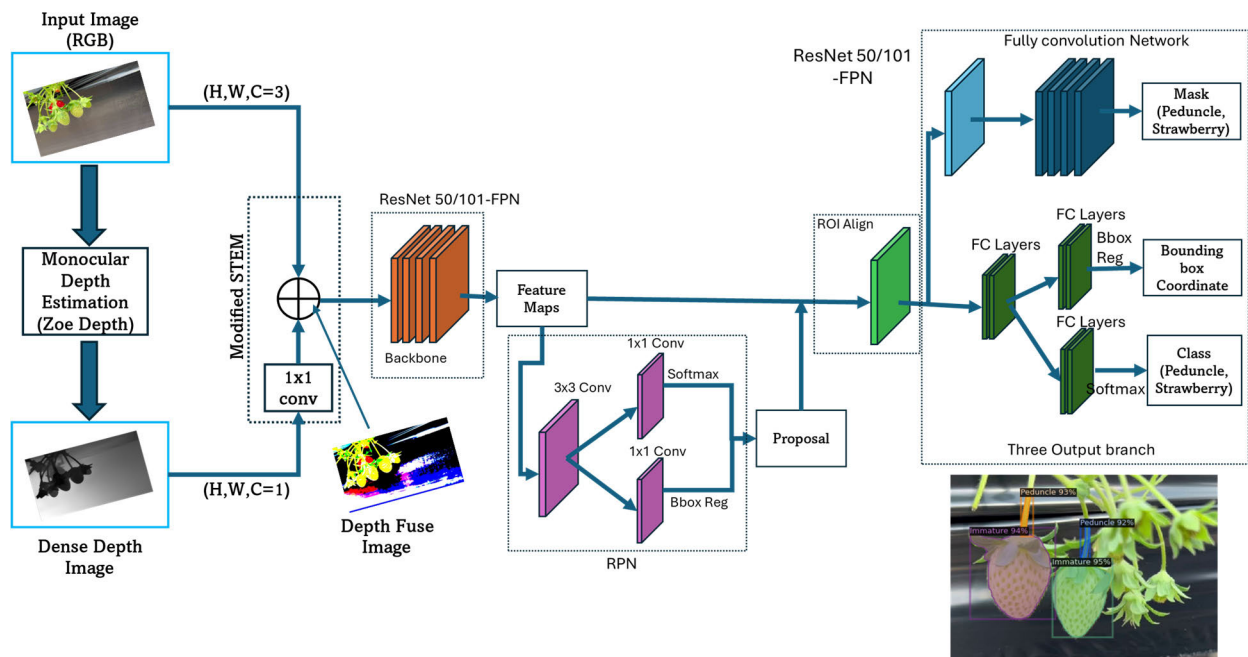


FIGURE 6. Detailed schematic diagram of DF-Mask R-CNN with Monocular depth estimation, ZoeDepth.

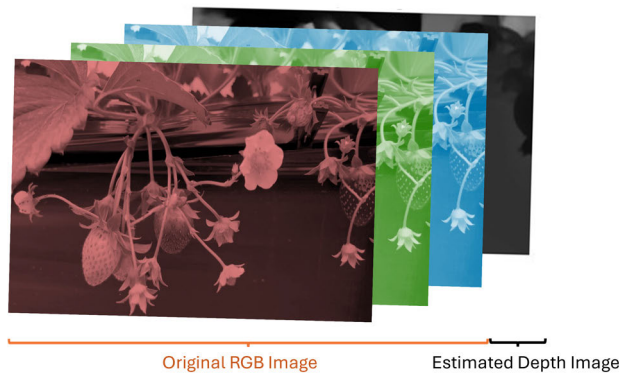


FIGURE 7. 4-Channel RGB image with embedded depth channel obtained from monocular depth estimator.

E. IMPROVING INSTANCE SEGMENTATION THROUGH MONOCULAR DEPTH ESTIMATION

Mask R-CNN utilized RGB images as input for detection and instance segmentation of strawberries and corresponding peduncles. The RGB images offer texture and colour details of the ripe strawberry to distinguish fruits from their background, while elements like unripe strawberries and peduncles often share similar visual characteristics with branches and leaves, complicating their detection. Additionally, multiple fruit adhesions and overlapping cause difficulty distinguishing fruits. Monocular depth estimation, which derives depth information from a single RGB image through convolutional neural networks (CNNs), provides pixel-wise spatial variation within the scene objects. The results of the model for the generated dense depth map are shown

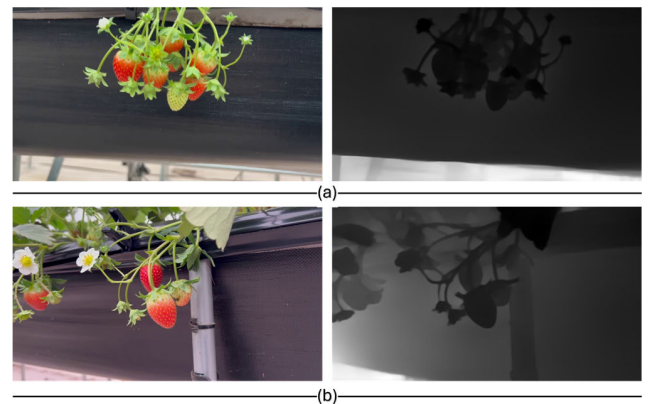


FIGURE 8. Depth map obtained from ZoeDepth monocular depth estimator (a) dense depth map of strawberry in dense fruitlet, (b) dense depth map of strawberry under occlusion.

in Figure 8. The figure suggested that the estimated depth map offers increasing contrast between elements of the scene foreground and background. Therefore, incorporating depth information as an additional channel in Mask R-CNN provides depth cues to the instance segmentation model.

F. DEPTH ESTIMATION OF PICKING POINT LOCATION ON THE PEDUNCLE

To estimate the optimal picking point for bruise and damage-free harvesting of ripe strawberries, two considerations were made: one ripe strawberry detection confidence threshold was set greater than 0.7, and other picking point location lay between 13 mm to 20 mm above the strawberry

calyx on the peduncle [6], [8]. In this study, 15 mm was taken as the reference picking point. Besides the distance from the peduncle base, it was crucial to consider the depth value at the picking point with reference to the camera. To ascertain the depth value, instance segmentation of the strawberry and corresponding peduncle was combined with depth data obtained from a LiDAR sensor. This combination facilitated the estimated depth values for the detected ripe strawberry fruit and its peduncle.

The masked depth value pixel information was refined, removing outliers by excluding depth value pixels outside the 25 cm to 1 m range. This step helped eliminate background-induced noises. The refined depth values for the strawberry fruit were used to determine the fruit's centroid, while for the peduncle, the mean depth value was calculated based on the closest 10% depth values in the peduncle segment.

The picking point detection was based on two scenarios, depending on the visibility of the peduncle associated with ripe strawberries. If the peduncle was visible and detected, the picking point was calculated directly else if not visible, the contour of the strawberry was used to estimate the peduncle's location and, consequently, determine the picking point. In this research, the estimation of the picking point when the peduncle was obscured was based on the strawberry's geometry. The method includes determining the principal axis and centroid of strawberry, then evaluating the peduncle and picking point estimation.

(a) The binary mask of the ripe strawberry was obtained using DF-Mask R-CNN. The contour of the strawberry was then used to calculate the centroid of the strawberry. Next, the principal axis was determined as the line passing through the centroid, representing the direction of maximum variance in the strawberry's shape.

(b) The intersection of the principal axis with the strawberry contour on the side farthest from the centroid was estimated as the location of the peduncle. The picking point was then determined based on this estimated peduncle location.

To assess the accuracy of depth measurements at the picking point, the estimated values were compared against ground truth values. These ground truth depths were obtained using a TF Luna 8m laser meter, which measured the distance from the RGB image-acquiring camera to the picking point on the peduncle. This laser meter could measure distances ranging from 10 cm to 800 cm. Approximately 300 picking point samples on the strawberry peduncles were analyzed to validate the method's accuracy.

G. EXPERIMENTAL PLATFORM AND PARAMETER SETTING

The research was conducted on a computer with the following specifications: an Intel Core i7-10700 CPU @ 2.90 GHz, 16 GB RAM, and an NVIDIA GeForce RTX 2060 graphic card with 6 GB dedicated memory. The operating system used was Microsoft Windows 10 Pro. The study used the Pytorch framework, version 2.3.0, with CUDA 12.1 and Cudnn 8.8 to leverage GPU acceleration for model training and inference.

The model training parameters used weight initialization with pre-trained weights from Resnet50-FPN and ResNet101-FPN backbone for RGB instance segmentation. Data augmentation techniques, including horizontal flipping with a probability of 0.5 and random scaling (ranging from 0.8 to 1.2), were applied to improve model accuracy by improving generalization and reducing overfitting. Training hyperparameters were carefully optimized: the learning rate is 0.00025, batch size per image equals 1, and the number of Region of Interest (ROIs) per batch equals 512. For inference, the ROI head Non-Maximum Suppression (NMS) threshold was configured at 0.1, with a score threshold of 0.7.

The experiment facilitated the training and fine-tuning of the RGB Mask R-CNN model using transfer learning to expedite the process. In contrast, the DF-Mask R-CNN, utilizing a 4-channel input, was trained from scratch. Across all models, the iteration batch size was maintained at 1, with a default decay coefficient of 0.0001. The total number of iterations was set at 30,000. The training time for Mask R-CNN and DF-Mask R-CNN was 3 hours 54 minutes and 3 hours 24 minutes, respectively. As shown in Figure 9, the accuracy graph for Mask R-CNN accuracy graph starts at 0.8, while the graph for DF-Mask R-CNN starts at 0.6. Both models eventually converge to an accuracy of approximately 0.96 after 30K iterations. The difference in the starting accuracy between Mask R-CNN and DF-Mask R-CNN can be attributed to the effects of transfer learning. Similarly, the total loss deviation decreases with each iteration and stabilises at 0.4 at the end of the training process for Mask R-CNN and DF-Mask R-CNN.

H. EVALUATION METRICS

The experiment uses standard metrics: Precision (P), Recall (R), F1-score, and mean average precision (mAP) to evaluate the performance of bounding box object detection, while the instance segmentation model performance was evaluated on the criterion of Microsoft COCO. This utilizes mean average precision (mAP) as the primary metric and evaluates mAP for small (mAP_{small}), medium (mAP_{medium}) and large (mAP_{large}) size target objects. Similarly, based on intersection over union (IOU) mAP values for 0.50 ($mAP_{IOU}=0.5$), 0.75 ($mAP_{IOU}=0.75$) and 0.5-0.95 ($mAP_{IOU}=0.50-0.95$). Although the model size and inference time are important parameters, they were not considered for the present study, and only inference time for the strawberry classes will be evaluated in terms of frame per second (fps).

$$Precision = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \times 100\% \quad (1)$$

$$Recall = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \times 100\% \quad (2)$$

$$F1 - Score = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100\% \quad (3)$$

$$MAP = \frac{\sum_{k=1}^N \text{Average Precision of } k\text{th class}}{\text{Total number of } k \text{ class (N)}} \quad (4)$$

Precision and recall evaluate the model's ability to detect and locate fruit classes accurately. The F1 score offers a balanced evaluation of the model's classification performance by combining precision and recall. Furthermore, mAP@0.5 measures the average precision across various confidence levels for predicted bounding boxes using an IOU threshold greater than 0.5 compared to the ground truth. In equations (1)-(3), the true positive represents the correctly detected strawberry and peduncle classes, the false positive represents the falsely detected strawberry and peduncle classes, and the false negative accounts for the number of missed classes detected.

The performance of estimating the picking point on the peduncle estimation was evaluated using example datasets. For each peduncle corresponding to ripe strawberries, depth estimation was conducted using both direct method and analysis based on the strawberry size. The accuracy of this depth estimation was assessed by calculating the root mean square error (RMSE) and mean absolute error (MAE).

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - x_{gt,i})^2} \quad (5)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |x - x_{gt,i}| \quad (6)$$

where,

x_i = distance estimation of i^{th} sample picking point on peduncle

$x_{gt,i}$ = ground truth distance of i^{th} sample picking point on peduncle

N = number of samples

III. RESULTS

In this study, to assess the effectiveness of the DF-Mask R-CNN, its performance was compared with the Mask R-CNN model. ResNet50-FPN and ResNet101-FPN backbone were used to explore the feature extraction capability of each model. The standard evaluation metrics were used for quantitative analysis on bounding box performance and instance segmentation performance of models. Apart from the quantitative analysis, the models were also compared based on qualitative by observing the performance of the strawberry and peduncle detection on the inference result. Furthermore, depth estimation accuracy for the picking point on the peduncle of ripe strawberries was evaluated.

A. OVERALL BOUNDING BOX DETECTION AND INSTANCE SEGMENTATION PERFORMANCE

Table 3 provides an overview of the bounding box detection performance of the Mask R-CNN and DF-Mask R-CNN models. The DF-Mask R-CNN with a ResNet101-FPN backbone achieved the highest performance among all models, with precision, recall, F1-score, and mAP values of 89.6%, 94.6%, 92.4%, and 99.3%, respectively. Mask R-CNN with ResNet101-FPN exhibited a precision

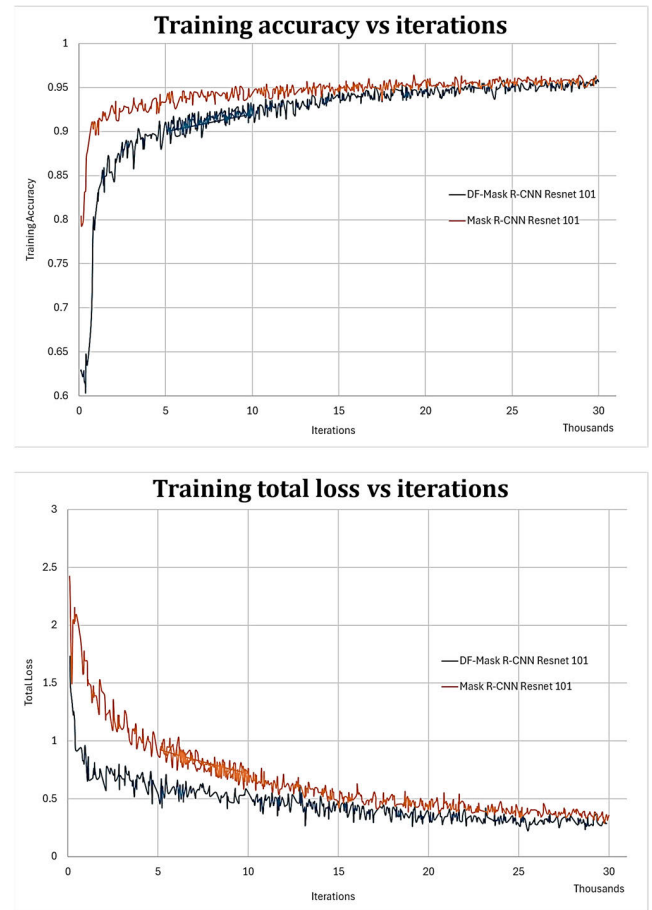


FIGURE 9. Training curves of accuracy and total loss function for DF-Mask R-CNN and Mask R-CNN ResNet101-FPN.

TABLE 3. Bounding box performance result for the best model trained (Mask R-CNN and DF-Mask R-CNN), obtained for the validation and test subsets (confidence score threshold set: 0.7).

Mask RCNN Model	Precision (%)	Recall (%)	F1-Score (%)	mAP@0.5 (%)
Mask R-CNN (ResNet50-FPN)	83.22	82.71	82.96	96.24
Mask R-CNN (ResNet101-FPN)	90.50	93.40	91.93	95.60
DF-Mask R-CNN (ResNet50-FPN)	81.62	87.72	84.56	98.13
DF-Mask R-CNN (ResNet101-FPN)	89.60	94.60	92.40	99.30

of 90.5%, which, while slightly higher than that of the DF-Mask R-CNN, remains comparable. Both models demonstrated progressive improvements when transitioning from the ResNet50-FPN to the ResNet101-FPN backbone.

Table 4 details the instance segmentation performance of Mask R-CNN and DF-Mask R-CNN, providing a comprehensive analysis of overall mAP and mAP as per object class

TABLE 4. Instance segmentation performance result for the best model trained (Mask R-CNN and DF-Mask R-CNN), obtained for validation and test subsets (confidence score threshold: 0.7).

Metric/ Model	Average Precision (AP, %)			
	Mask R-CNN (ResNet50-FPN)	Mask R-CNN (ResNet101-FPN)	DF-Mask R-CNN (ResNet50-FPN)	DF-Mask R-CNN (ResNet101-FPN)
mAP	71.5	75.8	73.3	81.9
mAP _{small}	12.0	21.3	27.5	33.3
mAP _{medium}	76.5	80.3	68.9	78.7
mAP _{large}	81.2	84.6	81.6	88.8
AP ^{IOU=0.50}	94.9	95.6	97.3	98.1
AP ^{IOU=0.75}	78.7	84.6	87.7	94.5

pixel sizes and different levels of overlap criteria. DF-Mask R-CNN with the ResNet101-FPN backbone outperformed all models, achieving an overall mAP of 81.9%, with mAP_{small}, mAP_{medium}, and mAP_{large} having values of 33.3%, 78.7% and 88.8%, respectively and an AP^{IOU=0.50} of 98.1%. Transitioning from the ResNet50-FPN to the ResNet101-FPN backbone, the DF-Mask R-CNN significantly boosted overall, class pixel size, and level of overlap criteria-specific mAP segmentation performance.

B. CLASS-WISE BOUNDING BOX DETECTION AND INSTANCE SEGMENTATION PERFORMANCE

Table 5 showcases the class-wise bounding box performance of Mask R-CNN and DF-Mask R-CNN models with different backbones. Using a ResNet101-FPN backbone, the DF-Mask R-CNN excels in detecting transient strawberry and peduncle classes, achieving AP scores of 96.82% and 84.78%, respectively. In contrast, the Mask R-CNN with the same backbone performs best in immature and mature strawberry classes, with AP scores of 90.25% and 91.29%. Overall, the ResNet101-FPN backbone consistently outperforms the ResNet50-FPN backbone across both models.

Table 6 illustrates the class-wise instance segmentation performance for Mask R-CNN and DF-Mask R-CNN models with ResNet50-FPN and ResNet100-FPN backbones. The DF-Mask R-CNN with a ResNet101-FPN backbone delivers the highest performance in class-wise segmentation, achieving AP scores of 86.99% for immature, 85.14% for mature, 90.74% for transient, and 64.83% for peduncle classes. Similarly, the Mask R-CNN with ResNet101-FPN achieves AP scores of 83.98% for immature, 82.51% for mature, 81.36% for transient, and 55.44% for peduncle classes.

C. PERFORMANCE ON DETECTION OF STRAWBERRY AND PEDUNCLE

In a clearly visible environment, the detection performance of Mask R-CNN closely mirrors that of DF-Mask R-CNN for identification and segmentation, as illustrated in Figure 10(a). Both models accurately identify strawberries without errors

TABLE 5. Class-wise bounding box performance for the best model trained (Mask R-CNN and DF-Mask R-CNN), obtained for the validation and test subsets (confidence score threshold set to 0.7).

Mask RCNN Model	Average Precision (AP, %)			
	Immature	Mature	Transient	Peduncle
Mask R-CNN (ResNet50-FPN)	84.84	88.46	83.69	75.90
Mask R-CNN (ResNet101-FPN)	90.25	91.29	86.19	80.59
DF-Mask R-CNN (ResNet50-FPN)	83.51	80.58	91.66	70.77
DF-Mask R-CNN (ResNet101-FPN)	90.25	87.02	96.82	84.78

TABLE 6. Class-wise instance segmentation performance for the best model trained (Mask R-CNN and DF-Mask R-CNN), obtained for the validation and test subsets (Confidence score threshold: 0.7).

Mask RCNN Model	Average Precision (AP, %)			
	Immature	Mature	Transient	Peduncle
Mask R-CNN (ResNet50-FPN)	82.43	82.56	81.05	53.19
Mask R-CNN (ResNet101-FPN)	83.98	82.51	81.36	55.44
DF-Mask R-CNN (ResNet50-FPN)	80.25	77.85	83.89	51.14
DF-Mask R-CNN (ResNet101-FPN)	86.99	85.14	90.74	64.83

or omission, demonstrating relatively high performance. However, in a cluttered and occluded environment, the performance of DF-Mask R-CNN was better, as shown in Figure 10(b) – (d). In Figure 10(b), Mask R-CNN misses unripe strawberries and misidentifies transient class strawberries as mature. In Figure 10(c), Mask R-CNN misses unripe fruit and its peduncle. Similarly, in Figure 10(d), DF-Mask R-CNN performs better instance segmentation of strawberries and associate peduncles than Mask-RCNN.

D. DEPTH ESTIMATION

The performance on depth estimation for picking points on the available peduncle mask exhibits a mean absolute error of 1.98 cm and an RMSE of 2.13 cm.

IV. DISCUSSION

A. MODEL ARCHITECTURE

In response to the challenges of difficulty in detecting the thin structure peduncle and strawberry under branching, overlapping and occlusion, Mask R-CNN architecture had been modified to introduce a new dense depth channel along with RGB image to create DF-Mask R-CNN. This provides spatial information about the target object and different elements

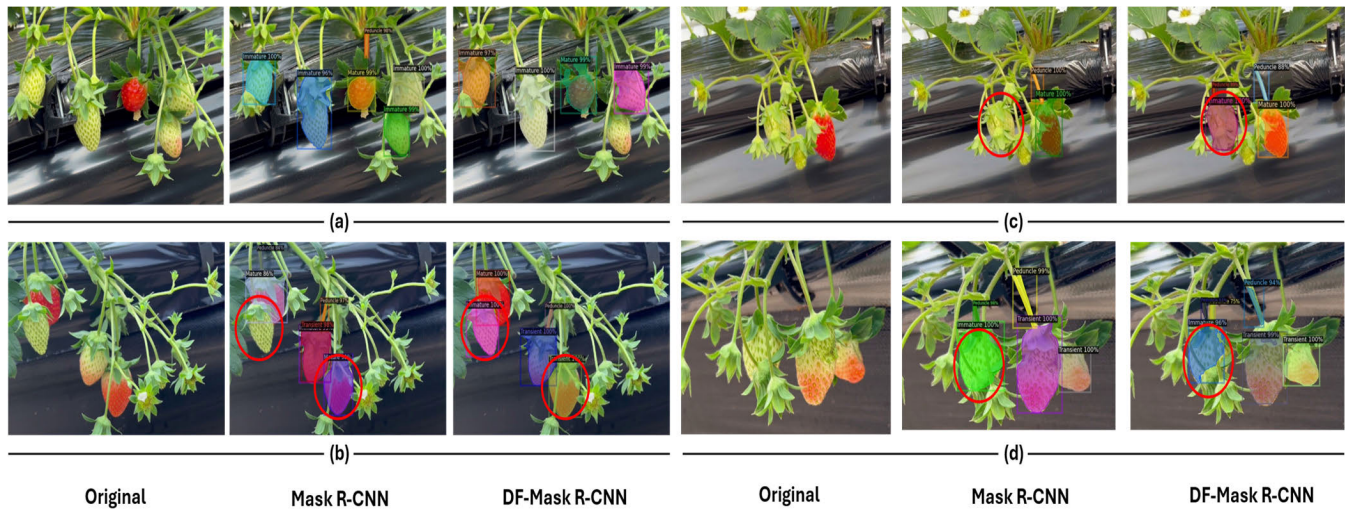


FIGURE 10. Qualitative analysis on detection of strawberry fruit and peduncle with Mask R-CNN and DF-Mask R-CNN model (a) Fruits with the clear difference (b) Missed and wrong classed fruit (c) Fruit and peduncle under occlusion and heavy branching (d) Segmentation.

in the scene, supplementing color and texture information of the RGB image to enhance identification and improve the instance segmentation. The performance of the DF-Mask R-CNN attributes promising results and further exploration for autonomous harvesting.

B. ISSUES ON IDENTIFICATION OF STRAWBERRY AND PEDUNCLE

This study used DF-Mask R-CNN to enhance the detection and segmentation of the strawberry and its peduncle. The segmentation results in Figure 11 illustrates the original image and detection of strawberries and peduncles using Mask R-CNN and DF-Mask R-CNN. The results with a red circle denote the inconsistencies in the identification and segmentation of the strawberries and their peduncle by Mask R-CNN. The figure shows the condition of missed fruits due to small size and severe occlusion. Similarly, the missing peduncle was observed due to a cluttered environment, a background with a similar color, and occlusion due to the leaf. Other reasons were wrongly detected peduncles, missed peduncles due to strawberry pose, and fruit occlusion associated with fruit. DF-Mask R-CNN addressed these discrepancies. The improvement could be attributed to the dense depth information fused into the RGB image, which allows for distinguishing different elements such as fruit, peduncle, leaf, branch, and background. Additionally, the better association observed between the strawberry and its peduncle detection could be because of the annotation rules that led the segmented mask of the strawberry and its peduncle to overlap for association.

Considering the ResNet101-FPN backbone of DF-Mask R-CNN and Mask R-CNN, the models were compared based on overall bounding box identification and instance segmentation performance. Bounding box performance for both models showed good strawberry and peduncle identification, with F1-scores of 91.93% and 92.40% for Mask R-CNN

and DF-Mask R-CNN, respectively. However, regarding localization identification capability, DF-Mask R-CNN outperformed Mask R-CNN by 3.7% with a $mAP@0.5$ of 99.3%. Similarly, in the instance segmentation performance comparison between DF-Mask R-CNN and Mask R-CNN, the metric differences for mAP , mAP_{small} , and $AP^{IOU=0.5}$ were 6.1%, 12%, and 2.5%, respectively. This suggests a significant improvement in mAP_{small} , indicating better segmentation for small pixel objects by the DF-Mask R-CNN algorithm.

Overall, DF-Mask R-CNN and Mask R-CNN have good detection, localization performance, and instance segmentation capabilities for various classes of strawberries. However, Mask R-CNN struggles to correctly detect small strawberry fruits, thin peduncles, and multiple classes of fruits under cluttered environments.

In class-wise segmentation performance comparison between Mask R-CNN and DF-Mask R-CNN with ResNet101-FPN, for immature and mature classes, only a considerable increment of approximately 3% was observed between the models. In contrast, a significant increment of 8.86% was observed for the peduncle class, demonstrating better detection of thin-structured peduncles by DF-Mask R-CNN.

In the overall assessment of DF-Mask R-CNN over Mask R-CNN, DF-Mask R-CNN demonstrates superior performance in detecting both strawberries and peduncles, correctly identifying the peduncle associated with strawberries, providing better confidence scores, and offering more consistent detection quality across different scenarios.

C. ISSUES ON THE PICKING POINT ESTIMATION IDENTIFICATION

The peduncle of the strawberry needs to be determined to estimate the picking point for autonomous strawberry harvesting. The peduncle of the strawberry could be either



FIGURE 11. Comparison of the strawberry fruit and peduncle segmentation with Mask R-CNN and DF-Mask R-CNN (a) Missed small strawberry fruit (b) Heavy occluded fruit detected (c) Missed peduncle detection (d) Missed peduncle due to the similar background (e) Missing peduncle due to leaf occlusion (f) Wrongly detected peduncle associated with strawberry (g) Missed peduncle due to ill pose strawberry (h) Missed peduncle detection of occluded strawberry.

directly detected or estimated based on the shape and orientation of the strawberry. In a study by Yu et al. [6], [8], the strawberries detected with more than 90% visibility were treated as correctly detected strawberries. Then, the contour of the strawberry was used to estimate the peduncle and, in turn, the picking point. Further, they developed R-YOLO, which determines the pose of the strawberry, which increases the accuracy of determination of the picking point, increasing the harvesting success rate up to 84.35%. The limitation of the study was that strawberries need to be fully visible; medium and severely occluded strawberries were not detected and harvested. Similarly, the research by Lemsalu et al. [30] explored the detection of the peduncle associated with the ripe strawberry in ground-based cultivation by taking images from 10 to 50 cm from the strawberry fruit. The vision system used the YOLO algorithm and bounding box detection. However, the research suffered proper visibility of peduncles from a distance and failed to detect the correct peduncles under densely clustered berries.

Our study utilized DF-Mask R-CNN to detect the peduncle and perform instance segmentation directly. This allows the depth estimation and localization of the picking point on the peduncle for associated ripe strawberries. Similarly, in the absence of the peduncle visibility, due to the high-performance instance segmentation capability of the model, the strawberry contour could be accurately estimated, and better localization of the peduncle and picking point could be achieved. Therefore, DF-Mask RCNN can significantly improve the detection and localization of fruit

attributes in the on-field greenhouse, increasing the efficiency of autonomous robots.

D. COMPARISON WITH RELATED METHODS

Our model mainly focused on enhancing the detection of small strawberries and thin structure peduncles. To achieve this, we analyzed several existing studies on strawberry detection and localization models for comparison using bounding box performance: SDNet. [15], DSE-YOLO [31], YOLOv5s [12], YOLOv5s6 [12]. Even though the models' architecture and the field environment used in those studies varied, it was challenging to attain the same level of accuracy on our dataset. Nonetheless, we analyze the models based on accuracy and localization. Table 7 shows the performance metrics F1-score and $\text{map}@0.5$ of different models with our model DF-Mask R-CNN ResNet101-FPN. Among all the compared models, our model outperforms all the models with an F1-score of 92.4% and $\text{mAP}@0.5$ of 99.3%. The model that came close to our model was SDNet, with an F1-score of 91.9% and $\text{mAP}@0.5$ of 94.26%. Similarly, in the case of the segmentation performance study on Mask R-CNN by Pérez-Borrero et al. [32] had mAP of 43.85%, $\text{mAP}_{\text{small}}$ of 7.54%, and $\text{AP}^{\text{IOU}=0.5}$ of 74.2% which are significantly less than our model with mAP of 81.9%, $\text{mAP}_{\text{small}}$ of 33.3% and $\text{AP}^{\text{IOU}=0.5}$ of 98.1% suggesting best accuracy on detection and segmentation of the strawberry fruit and its peduncle. The limitation related to our model is that the inference time it took was 0.31 sec, which hinders the real-time processing.

TABLE 7. Comparison with existing detection models.

Models	F1-score (%)	mAP@0.5
SDNet	91.91	94.26
DSE-YOLO	82	87
YOLOv5s	86.6	90.7
YOLOv5s6	86.6	91.1
DF-Mask R-CNN(ResNet101-FPN)	92.4	99.3

V. CONCLUSION

This paper introduces an innovative methodology to enhance the precise detection and localization of the picking point on the peduncle of ripe strawberries, addressing challenges such as the thin structure of peduncles, dependencies on strawberry morphology, and occlusions in unstructured environments. The proposed approach, termed DF-Mask R-CNN (Depth-Fused Mask R-CNN), leverages a ResNet101-FPN backbone for improved instance segmentation and integrates dense depth information generated by ZoeDepth, a state-of-the-art monocular depth estimation model. This fusion of depth data with instance segmentation enables accurate 3D localization of the picking point, overcoming limitations posed by traditional methods. The DF-Mask R-CNN was tested on a prepared dataset, achieving outstanding results. It recorded the highest bounding box performance with an F1-score of 92.4% and mAP of 99.3%. For segmentation performance, it achieved an overall mAP of 81.9%, with mAP_{small} at 33.3%, mAP_{medium} at 78.79%, mAP_{large} at 88.8 and AP^{IOU=0.5} at 98.1%. The model's inference time for detection and localization was 0.31 seconds, demonstrating its efficiency in real-time applications. The comprehensive performance evaluation of the DF-Mask R-CNN exhibits its capability to precisely detect peduncle segments, overcoming the challenges posed by peduncle segmentation, strawberry morphology, and occlusions in unstructured environments. It ensures precise depth estimation of the picking points and excels in segregating strawberry fruits under complex conditions like multi-adhesion, overlapping, and occlusion.

Future enhancements will focus on improving the robustness and accuracy of the monocular depth estimation, enabling direct depth measurement without relying on traditional depth sensors. Additionally, efforts will be directed towards streamlining the DF-Mask R-CNN to optimize its efficiency while maintaining high performance.

DATA AVAILABILITY

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CONFLICTS OF INTEREST

On behalf of all the authors, the corresponding author declares, there is no conflict of interest.

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